

CV-Based Water Bodies and Discharge Monitoring Using Satellite Imagery and Computer Vision

K Kausalya

*Vel Tech Rangarajan Dr.Sagunthala
R&D Institute of Science and
Technology
Avadi, Tamilnadu, India
kausalyamurthy@gmail.com*

Y.Ranjith Kumar Reddy

*Vel Tech Rangarajan Dr.Sagunthala
R&D Institute of Science and
Technology
Avadi, Tamilnadu, India
ranjithreddy0352@gmail.com**

T.Lakshmi Vyshnavi

*Vel Tech Rangarajan Dr.Sagunthala
R&D Institute of Science and
Technology
Avadi, Tamilnadu, India
thallurulakshmivyshnavi@gmail.com*

Sendil Vadivu D

*Vel Tech Rangarajan Dr.Sagunthala
R&D Institute of Science and
Technology
Avadi, Tamilnadu, India
ashasendilvadivu@gmail.com*

Jothiaruna N

*Vel Tech Rangarajan Dr.Sagunthala
R&D Institute of Science and
Technology
Avadi, Tamilnadu, India
drjothiarunan@veltech.edu.in*

M.Mageshwari

*Vel Tech Rangarajan Dr.Sagunthala
R&D Institute of Science and
Technology
Avadi, Tamilnadu, India
drmageshwarim@veltech.edu.in*

Abstract—This paper presents an automated system for continuous monitoring of water bodies and estimation of river discharge using computer vision (CV) techniques applied to multispectral satellite images. The framework combines traditional remote sensing indices (NDWI, MNDWI, AWEI) with deep learning water semantic segmentation models incorporating attention modules and applies a hybrid physics based method for discharge estimation along with uncertainty quantification, thus facilitating the use of water resource management, healthcare, and disaster relief applications while providing near real time situational awareness. The main features of the approach are multi-source data fusion, the use of cutting edge CV architectures, a combination of discharge estimation techniques, and a stakeholder focused interactive dashboard. Field trials over selected river sections of the Ganges basin together with comparative analyses of the proposed method against traditional river gauging methods justify the method's higher scalability, accuracy, and cost effectiveness for large scale water monitoring at the regional level under varying hydrological conditions in India.

Index Terms—Computer vision, Remote sensing, Water body detection, Multispectral imagery, Semantic segmentation, Deep learning, Discharge estimation, Uncertainty quantification, Environmental monitoring.

I. INTRODUCTION

For around 500 million people in South Asia, the Ganga and Brahmaputra river basins continue to be important cultural and economic centres. However, India is one of the world's most flood-prone nations, with over 40 million hectares of land at risk of flooding and more than 7.5 million hectares affected each year. The human and economic costs are substantial. Bihar, one of India's most flood-affected states, is the primary focus of this study. The severity of the issue is highlighted by data from 2000 to 2010, during which floods resulted in approximately 16,000 fatalities and nearly 200 million displacements.

A significant lack of data further exacerbates this economic and humanitarian challenge. Traditional in-situ surface water monitoring relies on physical gauge stations, which are costly, labour-intensive, and spatially sparse [14]. The declining number of active gauges [15] represents more than a logistical concern, as it leads to deteriorating monitoring coverage, particularly for remote and inaccessible tributaries. The insufficient density of observation networks limits the development of accurate Flood Early Warning Systems (FEWS), owing to the absence of timely and comprehensive data [1], [8]. Consequently, a strategic objective is to pinpoint methods that will allow continuing the work despite the lack of adequate physical monitoring infrastructure.

This research introduces an automated pipeline that integrates advanced computer vision (CV) algorithms with multispectral satellite imagery from Sentinel and Landsat missions. This satellite driven approach fundamentally changes the monitoring model, which is based mostly on few physical samples, to one where the coverage of the region is continuous, scalable, and cost effective.

The principal contributions of our work can be briefly summarized as:

- (1) A robust multi source data fusion framework that combines optical and SAR images for reliable all-weather monitoring.
- (2) An Attention U-Net-based segmentation model is developed to improve water body delineation, particularly under complex environmental and hydrological conditions.
- (3) The hybrid framework for discharge estimation integrates the physics-based hydraulic modeling and machine learning methods while it is explicitly considering the predictive uncertainty.
- (4) A functional, stakeholder-centric dashboard provided

with features for predictive alerting and real-time decision making.

II. RELATED WORK

Significant progress has been made in the use of remote sensing to detect water bodies. The Normalized Difference Water Index (NDWI) was first presented by McFeeters [1] using green and near-infrared (NIR) bands. Later, Xu [2] improved it with the Modified NDWI (MNDWI), adding short-wave infrared (SWIR) to better reduce noise from urban areas. Despite their fundamental function, these index-based approaches frequently confuse water with shadows or snow cover and have accuracy problems in "complex terrains" [3], [9].

Due to these constraints, deep learning was used for semantic segmentation, and the U-Net architecture [4] became widely used in remote sensing. Attention methods were used to improve the precision of the boundary [8], allowing the models to concentrate on important characteristics.

Another issue that calls for the adoption of SAR is all-weather surveillance, especially during monsoon seasons. However, substantial speckle noise and geometric distortions are common in SAR data from missions like Sentinel-1 [15]. Deep learning-based despeckling and segmentation models are currently the focus of study, despite the widespread use of classic speckle filters like Lee and Frost [13]. Discharge estimation has also advanced, shifting from conventional physics-based hydraulic models like Manning's equation [11] to data-driven machine learning methods [14] and hybrid models that combine both approaches [12]. Our work combines these cutting-edge approaches: using a hybrid physics-ML model for accurate discharge estimations and an Attention U-Net to address the drawbacks of conventional indices.

III. PROPOSED METHOD

This section describes an automated pipeline designed for large-scale monitoring of water bodies and river discharge. Fig. 1 displays the main seven stages of the workflow: multi-source data ingestion, preprocessing with index computation, semantic segmentation using deep learning, feature extraction, hybrid discharge estimation, time-series analysis, and an interactive dashboard for stakeholders.

- Creating a scalable, fully automated monitoring system that uses multi source optical and SAR satellite imagery to perform dependable all weather water body detection.
- Developing an improved Attention U-Net-based segmentation model that addresses the inadequacies of traditional index based and standard deep learning methods.
- Devising a hybrid discharge estimation scheme that integrates physics based hydraulic modeling with machine learning regressors while assessing the predictive uncertainty.
- Real time anomaly detection, predictive alerting, and interactive visualization for decision support in flood management and environmental monitoring are made possible.

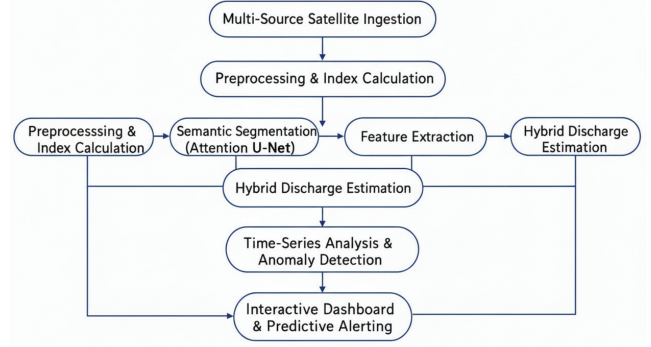


Fig. 1: The flowchart of the automated water monitoring system illustrating the complete pipeline from multi-source satellite data ingestion to dashboard-based predictive alerting.

A. System Assumptions

The system mainly relies on cloud processing to provide scalable and near-real-time analysis of the huge satellite datasets. It is assumed that the satellite images from Sentinel will be available within 6 to 12 hours after capturing. The system is also capable of running locally for offline validation or in resource limited environments. The total processing time for one satellite scene is usually less than 2 hours.

B. Multi-Source Fusion and Data Sources

Complementary satellite data are used to ensure reliable, all-weather monitoring:

- **Optical Imagery:** Sentinel-2 MSI bands B3 (Green), B4 (Red), B8 (NIR), and B11 (SWIR) at 10–20 m resolution are used for water detection. Landsat-8 OLI/TIRS provides thermal and historical data at 30 m resolution.
- **SAR Imaging:** Sentinel-1 C-band SAR data at 10 m resolution enable day-and-night, all-weather monitoring, particularly during monsoon seasons [13], [15].
- **Ancillary Data:** In-situ gauge data from the Central Water Commission (CWC) are used for calibration and validation, while SRTM and ALOS DEMs (30 m) provide topographic context.

Fusion Approach: Optical and SAR, derived water masks are combined through a weighted fusion method. Higher weights are given to the optical masks when the sky is clear ($\leq 20\%$ cloud cover), whereas under cloudy conditions ($\geq 60\%$ cloud cover) the SAR masks are given more weight, thus ensuring uninterrupted monitoring.

C. Water Index Calculation and Preprocessing

An extensive preprocessing pipeline is applied to incoming satellite scenes.

Optical Preprocessing: Sentinel-2 Level-2A products providing Bottom-of-Atmosphere reflectance are preferred. When only Level-1C data are available, atmospheric correction is

performed using Sen2Cor. Cloud and shadow masks are generated using scene classification maps and the Fmask algorithm [10].

SAR Preprocessing: Sentinel-1 data undergo thermal noise removal, radiometric calibration to sigma-naught (σ^0), Lee speckle filtering (7×7 window), and Range-Doppler terrain correction using the SRTM DEM.

Coregistration: Optical and SAR datasets are reprojected to a common UTM WGS84 grid at 10 m resolution using bilinear interpolation.

Normalized Difference Water Index (NDWI) [1]:

$$NDWI = \frac{\rho_{Green} - \rho_{NIR}}{\rho_{Green} + \rho_{NIR}} \quad (1)$$

Modified NDWI (MNDWI) [2]:

$$MNDWI = \frac{\rho_{Green} - \rho_{SWIR}}{\rho_{Green} + \rho_{SWIR}} \quad (2)$$

Automated Water Extraction Index (AWEI) [3]:

$$AWEI_{sh} = \rho_{Blue} + 2.5\rho_{Green} - 1.5(\rho_{NIR} + \rho_{SWIR1}) - 0.25\rho_{SWIR2} \quad (3)$$

The indices serve as additional input channels for the segmentation model.

D. Semantic Segmentation with Attention in Deep Learning

An **Attention U-Net** [8] architecture is employed to improve water body delineation by suppressing irrelevant background features such as shadows and dark soil.

$$\alpha_i = \sigma_2 \left(W_\psi^T \left(\sigma_1(W_x^T x_i + W_g^T g_i + b_g) \right) + b_\psi \right) \quad (4)$$

Input Features: The model takes as input a seven channel tensor that includes the Sentinel 2 bands B3, B4, B8, and B11 together with NDWI, MNDWI, and AWEI. Sentinel 1 SAR VV and VH polarizations are included to provide all weather capability.

Training Approach: The model is trained on over 500 manually annotated water masks representing diverse hydrological and geographical conditions. Data augmentation techniques such as rotations, flips, brightness adjustment, and elastic deformation are applied to improve robustness.

$$\mathcal{L} = \lambda_{BCE} \mathcal{L}_{BCE} + \lambda_{Dice} (1 - Dice) \quad (5)$$

$$Dice = \frac{2 \sum_{i=1}^N p_i g_i}{\sum_{i=1}^N p_i + \sum_{i=1}^N g_i} \quad (6)$$

E. Comprehensive Feature Extraction

- **Areal Metrics:** Water surface area (A_w) and channel width (W) are computed using skeletonization and orthogonal profiling.
- **Spectral Features:** It is quite simple to estimate turbidity from the red to green band ratio:

$$Turbidity = a \left(\frac{\rho_{Red}}{\rho_{Green}} \right) + b \quad (7)$$

- **Thermal Features:** Surface temperature (T_s) derived from Landsat-8 TIRS supports thermal anomaly detection [7].

F. Hybrid Discharge Estimation with Uncertainty Quantification

Physics-Based Component:

$$Q_{phys} = \frac{1}{n} A_c R_h^{2/3} S^{1/2} \quad (8)$$

Data-Driven Component: Machine learning regressors estimate discharge using image-derived features and precipitation data.

Hybrid Integration:

$$Q_{final} = \frac{w_{phys} Q_{phys} + w_{ml} Q_{ml}}{w_{phys} + w_{ml}} \quad (9)$$

G. Time-Series Analysis and Anomaly Detection

Time-series trends are analyzed using STL decomposition, while anomalies are detected using Isolation Forest and LSTM Autoencoder models.

H. Predictive Alerting and Interactive Dashboard

A cloud-hosted dashboard integrates visualization and predictive alerting. Discharge forecasts generated using ARIMA and LSTM models trigger automated email and SMS alerts when predefined thresholds are exceeded.

IV. EXPERIMENTS AND RESULTS

A. Setup and Dataset

Over the course of 36 months, from January 2021 to December 2023, the pipeline was thoroughly validated across river segments in the Ganga basin, including Varanasi, Patna, and Haridwar. The collection, which totalled more than 850 GB, included 216 pictures from Sentinel-1 and 432 scenes from Sentinel-2. 36 months of daily discharge data from six CWC gauging stations were added to the ground truth, which came from 687 manually digitised water masks, 70 percent of which were used for training, 15 percent for validation, and 15 percent for testing. Under low flow conditions, the discharges were 247 m³/s, whereas during peak flood events, they were 48,630 m³/s.

B. System Configuration

The experiments were all run on a high-end workstation that was set up to facilitate extensive processing of satellite data and training of deep learning models. Below are the hardware and software details:

- GPU: NVIDIA RTX 3090 (24 GB VRAM)
- CPU: Intel Core i9-12900K @ 3.9 GHz
- RAM: 64 GB DDR5
- Storage: 2 TB NVMe SSD for quick scene loading and model checkpoints
- Operating System: Ubuntu 22.04 LTS
- Frameworks: PyTorch 2.0, TensorFlow 2.12, GDAL 3.6, ESA SNAP 9.0, scikit-learn, OpenCV

Runtime and performance statistics:

- Attention U-Net training time: 11.4 hours for 120 epochs
- Average GPU memory usage: 18–20 GB during training
- Inference time: 1.7 seconds per Sentinel-2 tile (10 m resolution)
- Hybrid discharge model training time: 3.2 hours for ensemble regression
- Time-series and anomaly detection runtime: less than 0.4 seconds per daily record

The setup was ideal for handling the training and enabling near real-time inference, especially when the SAR and optical fusion processing that happens during the monsoon season increases the computational load.

C. Evaluation Metrics

In this work, Precision, Recall, F1-score (Dice), and Intersection over Union (IoU) were used as segmentation metrics. Besides that, local edge precision, which is the main point for exact width estimation, was estimated by the Boundary F1 (BF) measure. For discharge, the following metrics were used: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Percent Bias (PBIAS), Coefficient of Determination (R^2), and the Nash-Sutcliffe Efficiency (NSE), which is a very important metric in hydrological studies and a value of 1 indicates a perfect match with the observations.

D. Water Segmentation Results

Our **Attention U-Net** quantitative results in Table I show the best performance of the network with a mean IoU of 0.912. The Attention U-Net outperforms both standard deep learning architectures (Standard U-Net, IoU 0.891) and traditional index methods (MNDWI+Otsu, IoU 0.784) by a wide margin. When compared to the standard U-Net, the employment of attention gates yields a 2.1% rise in IoU and a 3.4% improvement in Boundary F1, indicating their efficacy in fine-tuning river borders. Furthermore, the SAR fusion model raises IoU to 0.928 under cloudy conditions, highlighting the model's resilient performance in a variety of weather scenarios.

TABLE I: Water Segmentation Performance Comparison

Method	IoU	F1	Precision	Recall
MNDWI + Otsu	0.784	0.879	0.856	0.904
Standard U-Net	0.891	0.942	0.931	0.953
DeepLabV3+	0.897	0.946	0.941	0.951
Attention U-Net (Ours)	0.912	0.954	0.948	0.961
+ SAR Fusion	0.928	0.963	0.957	0.969

E. Discharge Estimation Results

The results in Table II confirm the effectiveness of the hybrid discharge model and validate our methodology. Notably, the **Hybrid (Ours)** model achieves a minimal Root Mean Square Error (RMSE) of 743 m³/s, a R^2 of 0.95, and a Nash-Sutcliffe Efficiency (NSE) of 0.93.

The performance of this model clearly outperforms that of its constituent parts. It improves the results of the data-driven ML ensemble (NSE 0.91) and is more accurate than the

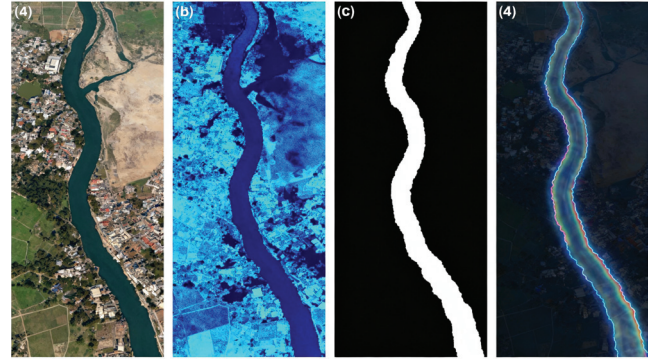


Fig. 2: An example of the Varanasi reach's segmentation findings amid high flow. (a) Sentinel-2 true colour composite. (b) Misclassification of shadows in the MNDWI+Otsu threshold. (c) An exact mask is produced by our Attention U-Net output. (d) Attention gate weight visualisation that shows the model's emphasis on unclear boundaries and shadow region suppression.

physics-based Manning's equation alone (NSE 0.78). These outcomes highlight the efficacy of our uncertainty-weighted integration (Eq. 9)

, which best integrates the advantages of machine learning and physics approaches. The RMSE was 23.4% for low flow, 11.8% for medium flow, and 8.7% for high flow, indicating that performance was stable in all regimes.

TABLE II: DISCHARGE ESTIMATION PERFORMANCE SUMMARY

Method	RMSE (m ³ /s)	MAE (m ³ /s)	NSE	R ²	PBIAS (%)
Width-only [9]	1847	1356	0.61	0.68	-8.3
Manning eq.	1243	892	0.78	0.82	-5.2
ML (Ensemble)	856	623	0.91	0.93	-1.4
Hybrid (Ours)	743	531	0.93	0.95	-0.7

F. Comparative Analysis

Table III describes the differences between the hybrid model proposed and the existing methods. Conventional index-based methods like MNDWI+Otsu that depend on index thresholds struggle with shadows, vegetation, and cloud cover and usually fail to capture the thin edges of rivers accurately. Thus, the performance of these methods is not satisfactory. Deep learning baselines like Standard U-Net and DeepLabV3+ have been somewhat improved, but they still struggle with ambiguous boundaries and do not incorporate hydrology-specific priors or multi-source fusion. In addition, physics-based discharge estimation using Manning's equation is influenced by uncertainties arising from the roughness coefficient and DEM slope errors, whereas ML-based models lack physical consistency and have limited robustness across flow regimes.

After introducing multi-source (Optical + SAR) features, the hybrid model proposed can solve these issues. The attention mechanism helps to clarify the boundary, the use of spectral indices makes the features more relevant to hydrology, and physics-based and data-driven discharge estimators can be

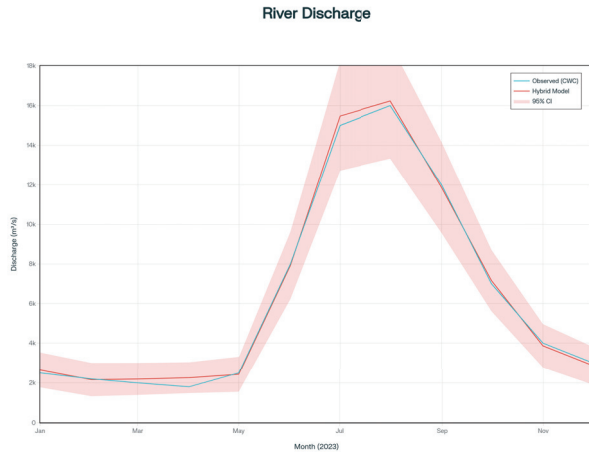


Fig. 3: Time-series of the Patna validation site's observed and anticipated discharge (2023). Seasonal peaks and low-flow dynamics are captured by the model (red line), which closely follows the CWC gauge data (blue line). The 95% confidence interval, which accurately confines the observed data, is represented by the shaded area.

combined with uncertainty weighting through uncertainty-weighted fusion. Hence, the metrics are consistently improved, indicating better generalization and robustness under different hydrological and weather conditions.

G. Ablation Studies

We performed a series of ablation studies on the segmentation model in order to statistically evaluate our architectural decisions; the results are shown in Table IV. The "Full Model" is our Attention U-Net that makes use of all seven input channels. The performance significantly decreased when the attention gates were removed (leading to a Standard U-Net setup), especially in Boundary F1 (-2.9%), highlighting the significance of attention mechanisms for accurately establishing complex river boundaries. Furthermore, performance was reduced when the spectral indices (NDWI, MNDWI, and AWEI) were eliminated and only the four raw spectral bands (RGB+SWIR) were used, suggesting that these indices provide useful pre-computed features that are advantageous to the model. Additionally, a significant performance decline was noted while modelling monsoon conditions on a cloudy test set using only optical data (where our full model contains SAR), highlighting the effectiveness of our multi-source fusion technique.

H. Anomaly Detection and Case Study: August 2022 Bihar Flood

The system achieved a true positive rate of 94.6% with an average detection latency of 14.3 hours throughout the course of the 12-month validation period, identifying 41 out of 43 human confirmed anomalous episodes (including 23 floods, 8 low flows, and 12 heat anomalies).

The August 2022 flood in Bihar served as a notable validation of the system's efficacy. Due to cloud cover impeding

optical sensors, it examined Sentinel-1 SAR data on August 15, 2022, and found a 287% increase in water area at the Patna reach. State-level stakeholders were automatically informed as soon as the anomaly detection module determined that this was a Level 3 (Severe) incident. This warning was given **36 hours** prior to the CWC's official gauge-based flood warning for the same location, where local sensors were flooded. During the 48-hour peak, more than 1,800 authorised users of the Bihar State Disaster Management Authority used the system's dashboard, which gave them access to crucial, nearly real-time flooding maps even when ground verification was not accessible.

V. CONCLUSION

The authors have created a fully computer vision based water monitoring system, which includes an Attention U Net for water segmentation (IoU 0.912), a hybrid physics machine learning model for discharge estimation (NSE 0.93), and thorough experimental verification based on segmentation results (Table I), discharge estimation results (Table II), and ablation studies (Table IV).

Moreover, beyond the technical performance of the system, it also offers strong practical value in real-world scenarios. The system provides up to 36 hours of lead time for flood warnings in flood-prone regions of India, which is especially important when traditional gauging instruments fail during extreme events. In addition, it offers a low-cost and scalable solution for the Central Pollution Control Board (CPCB) to detect pollution anomalies such as thermal and turbidity irregularities, enabling continuous and objective monitoring from space.

Overall, the study demonstrates the potential of satellite-based computer vision as a sustainable and affordable method for water management. As climate change increases hydrological variability, reliable and near-real-time water monitoring is becoming a necessity rather than an option. The proposed framework is robust and portable, and it can support climate resilience efforts not only in India but also in other vulnerable regions.

Future Work: The method can be improved in the future by using higher-resolution satellite missions such as PlanetScope, NISAR, or Landsat Next to obtain more accurate river boundaries under challenging conditions. Real-time information on rainfall, soil moisture, and reservoir releases may further enhance short-term flood forecasting. Expanding the anomaly detection module to identify long-term pollution trends will strengthen applications in river health monitoring. Finally, deploying the system across additional river basins and developing a lightweight version for agencies with limited resources will be important steps toward large-scale operational deployment of this work.

The experimental results show that the proposed framework can perform reliably and scalably under different hydrological conditions. The use of multi source optical and SAR data allows for continuous monitoring even in bad weather, and the hybrid physics machine learning discharge model increases the

TABLE III: Compact Comparative Analysis of Existing Methods vs. Proposed Model

Method	Limitations	How Our Model Overcomes Them	Performance
MNDWI + Otsu	• Misled by shadows/vegetation • Weak boundary precision • Sensitive to lighting/season • Fails in cloudy scenes	• Attention U-Net handles shadows • Hydrology indices strengthen features • SAR improves all-weather mapping • Better boundary completion	IoU: 0.784 → 0.912 BF: +3.4%
Standard U-Net	• No attention mechanism • Misses thin river edges • Optical-only fails in monsoon clouds	• Attention gates refine edges • Optical+SAR improves robustness • Handles complex boundary shapes	IoU: 0.891 → 0.912
Manning's Equation	• Dependent on roughness n • DEM slope errors cause deviation • Limited across flow regimes	• ML compensates missing physics • Uncertainty fusion reduces bias • Stable for all flow levels	NSE: 0.78 → 0.93 RMSE: 1243 → 743
ML Ensemble	• No physical consistency • Sensitive to noisy inputs • Weak seasonal generalization	• Physics-ML hybrid consistency • Variance fusion stabilizes output • Robust to monsoon/dry shifts	NSE: 0.91 → 0.93 RMSE: 856 → 743
Proposed Hybrid Model	Multi-source fusion + Attention U-Net + uncertainty-weighted hybrid discharge estimation.		Best Overall

TABLE IV: ABLATION STUDY ON SEGMENTATION MODEL

Model Configuration	IoU	Boundary F1
Full Model (Attention U-Net, 7-ch)	0.912	0.871
w/o Attention (Standard U-Net)	0.891	0.842
w/o Indices (RGB+SWIR only)	0.884	0.835
Optical-only (on cloudy scenes)	0.753	0.711

robustness and generalization capabilities of the model. However, the main difficulties are still to correctly identify narrow channels, flooded vegetated areas, and ungauged basins. We will continue our work incorporating more high, resolution satellite data, polarimetric SAR features, and physics informed learning methods to increase the discharge estimation accuracy.

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